# ■■ AI Code Review & Security Analysis Report

Automated Multi-Agent SAST & RAG Compliance Summary

| Total Findings | High Severity | Medium Severity | Low Severity |
|---|---|---|---|
| 6 | 3 | 3 | 0 |

## Detailed Findings & Remediation Roadmap

**#1 [High] Security - Line 9**
**Category:** OWASP-A07: Authentication Failures
**Issue:** Hardcoded sensitive credential or secret detected in source code.
**CWE/OWASP:** CWE-798 | A07:2025 Authentication Failures
**Remediation:** Hardcoding database credentials, API keys, or JSON Web Token (JWT) secrets directly into source code is a catastrophic security anti-pattern (frequently tracked under Bandit B105/B106). All sensitive configuration variables must be injected strictly via runtime environment variables or secure secret...

**#2 [High] Security - Line 10**
**Category:** OWASP-A07: Authentication Failures
**Issue:** Hardcoded sensitive credential or secret detected in source code.
**CWE/OWASP:** CWE-798 | A07:2025 Authentication Failures
**Remediation:** Hardcoding database credentials, API keys, or JSON Web Token (JWT) secrets directly into source code is a catastrophic security anti-pattern (frequently tracked under Bandit B105/B106). All sensitive configuration variables must be injected strictly via runtime environment variables or secure secret...

**#3 [High] Code Analysis - Line 13**
**Category:** Code Smell / Bug
**Issue:** String comparison using '==' operator instead of '.equals()'.
**CWE/OWASP:** CWE-476 | A10:2025 Mishandling of Exceptional Conditions
**Remediation:** Use .equals() to compare object contents rather than memory references. Appended Fix: Use string.equals() to compare object contents rather than references.

**#4 [Medium] Security - Line 21**
**Category:** OWASP-A04: Cryptographic Failures
**Issue:** Weak cryptographic hash algorithm or insecure block cipher mode (e.g., ECB/SHA-1) detected.
**CWE/OWASP:** CWE-328 | A04:2025 Cryptographic Failures
**Remediation:** Use strong cryptographic standards like SHA-256/SHA-3 and secure cipher modes like AES/GCM.

**#5 [Medium] Security - Line 22**
**Category:** OWASP-A04: Cryptographic Failures
**Issue:** Weak cryptographic hash algorithm or insecure block cipher mode (e.g., ECB/SHA-1) detected.
**CWE/OWASP:** CWE-328 | A04:2025 Cryptographic Failures
**Remediation:** Use strong cryptographic standards like SHA-256/SHA-3 and secure cipher modes like AES/GCM.

**#6 [Medium] Security - Line 28**
**Category:** OWASP-A04: Cryptographic Failures
**Issue:** Weak cryptographic hash algorithm or insecure block cipher mode (e.g., ECB/SHA-1) detected.
**CWE/OWASP:** CWE-328 | A04:2025 Cryptographic Failures
**Remediation:** Use strong cryptographic standards like SHA-256/SHA-3 and secure cipher modes like AES/GCM.